

Why simulate memory?

- Identify bottlenecks within memory bound hardware systems targeted towards high performance applications
- Over 50% of attention kernel cycles stalled due to DRAM access delays

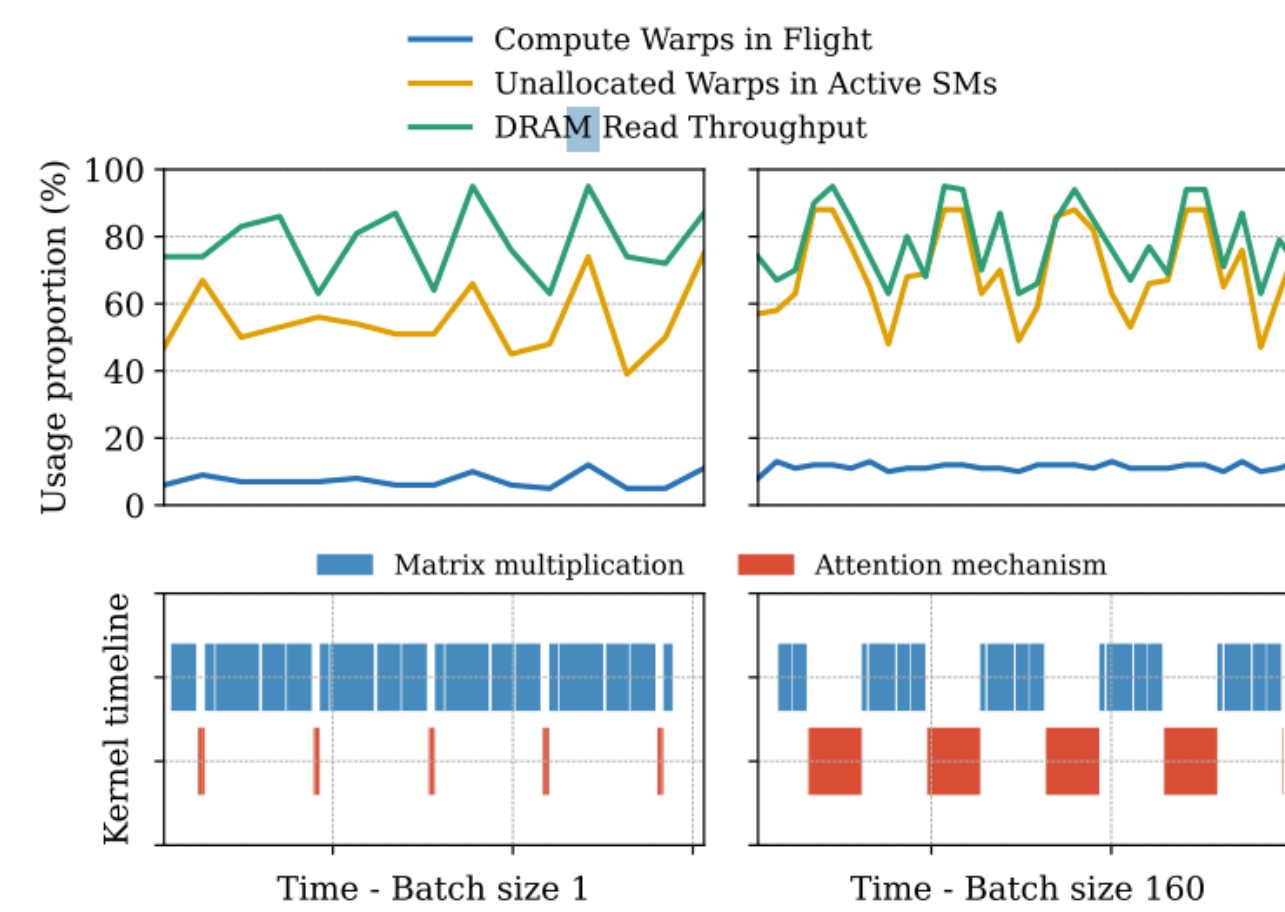


Figure 1: Utilization profiles of kernels during the first Llama-2-7B decode, ranging batches sizes 1 to 160. (IBM Research)

Current Infrastructure

- ML-Application Specific SoC Accelerators
 - Vector Accelerators
 - Deep Learning Accelerators
 - FFT Accelerators
 - Encryption Accelerators

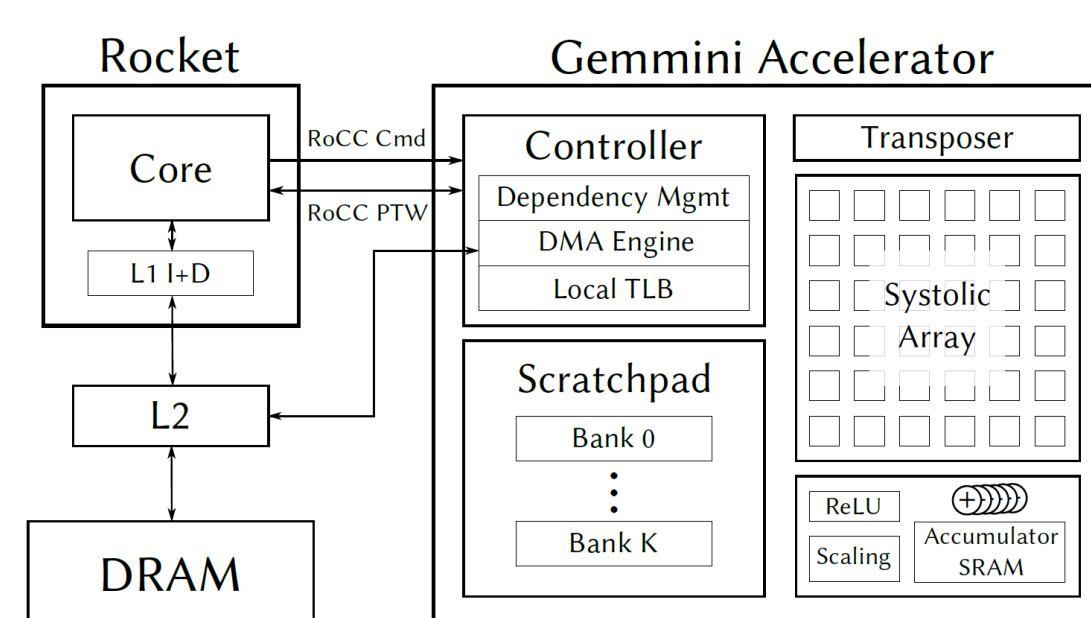
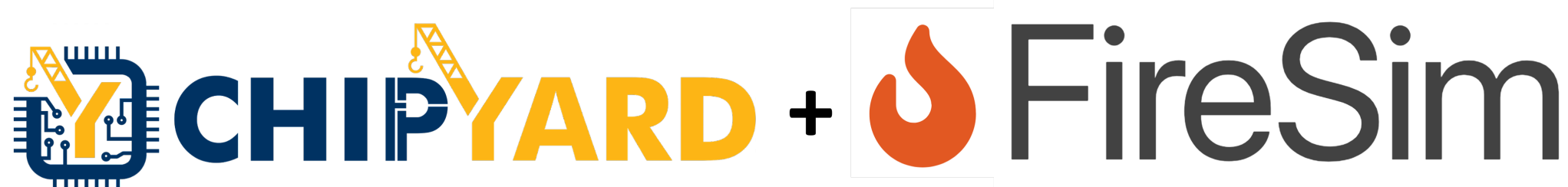


Figure 2: Gemini DNN Hardware Accelerator

- Chipyard + FireSim framework design-space exploration
 - Component placement (CPU-coupled, MMIO, etc)
 - Changing (i.e. larger register files, more accuracy, etc)
 - Add multiple devices of any type to the system
 - Implement and verify complex memory subsystem hierarchies



Project Goals

- How can we build a cycle-accurate, synthesizable RTL model of a memory subsystem to profile AI applications using FPGA emulation?
- How can we target specific memory subsystem hardware designs, such as HBM and GDDR5?

Initial Architecture Design

- Input Memory Command Trace for standalone evaluation
 - Valgrind + GCC for command trace extract from C program
 - Verilator for running RTL simulations
- Memory Controller
 - Bank-Level Parallelism
 - Same-row Transaction Level Optimization
 - Cache local read requests, and short-circuit read requests that access the same row
 - Invalidate write requests with subsequent write requests already in queue.
- Physical Memory Interface
 - Timing model based on previous-to-next transition timing matrix
 - Support for rank-level self refresh commands
 - Hierarchical Memory Orchestration
 - 8 channel, 2 ranks, and 8 bank configuration
 - Per banked queue distribution of requests and response coalescing
 - Bank level row profiling, and modeling of standard 1T DRAM HBM cell
- Trace-based behaviour
 - Able to characterize memory behaviour through command-level profiling, and bottlenecks through queue and bank level activity data.
 - Clearly quantify common memory controller level concerns such as request queue backpressure
 - Extensible (AXI-4 like), queue based frontend interface that can be injected into Chipyard SoC-to serve accelerator requests

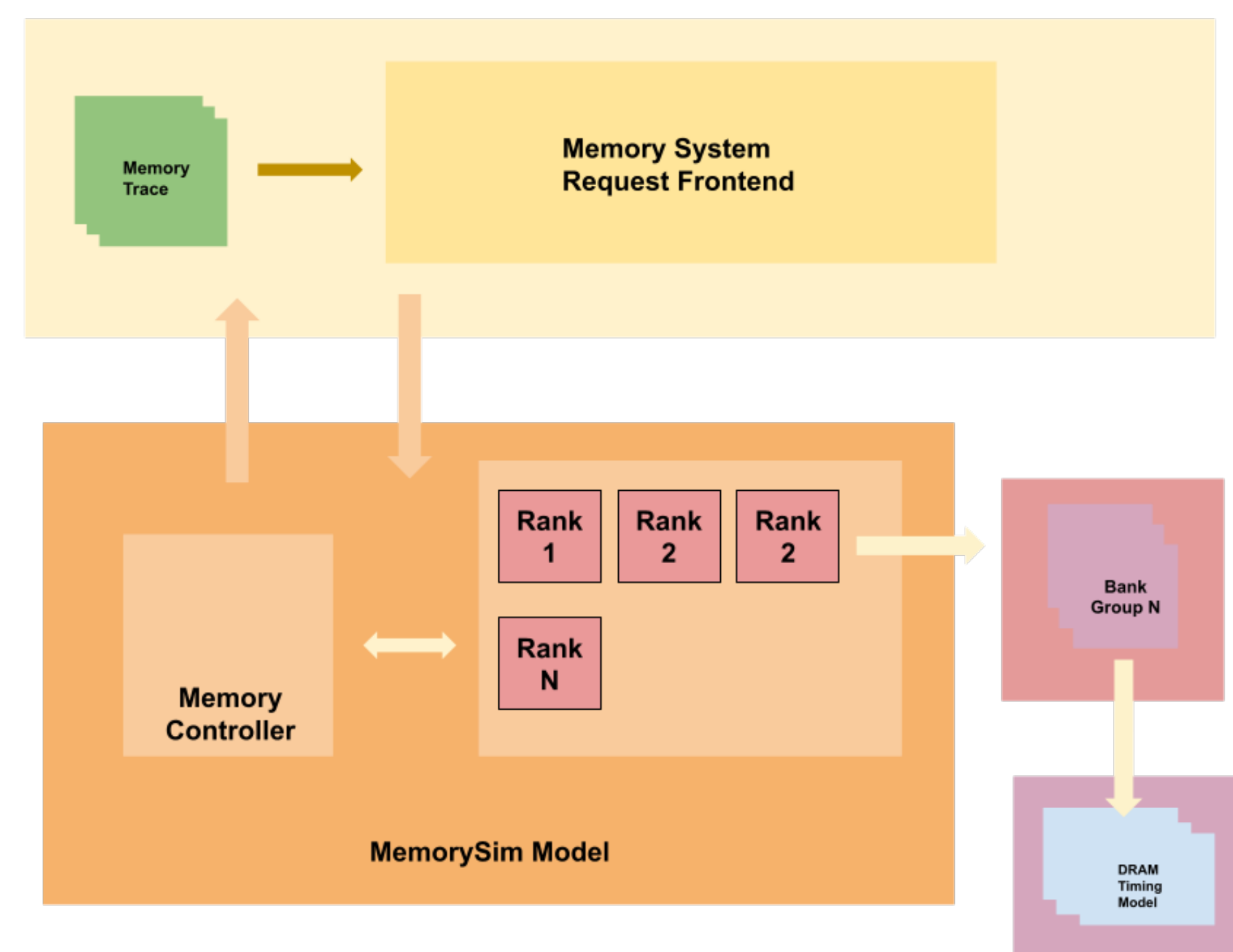


Figure 3: Memory Simulator High Level Architecture

Testing and Software

- Simulation setup
 - User timing parameters are defined based on memory subsystem specification
 - C++ simulation driver issues commands to memory subsystem model
 - Memory model returns responses, and **out** — **in** average latency is computed.
- Benchmarks are C-level programs implementing kernels for key AI applications
 - conv2d.c: A small 2D convolution kernel to evaluate spatial locality and burst access patterns in sliding-window operations.
 - multihead.attention.c: A toy multi-head attention workload that exercises dot-product computes and softmax-induced memory reuse typical of transformer models.
 - trace_example.c: A minimal read/write trace generator used to validate basic request sequencing and correct data return in the simulator.
 - vector.similarity.c: A cosine-similarity search across a small vector database to test irregular access patterns and reduction operations.

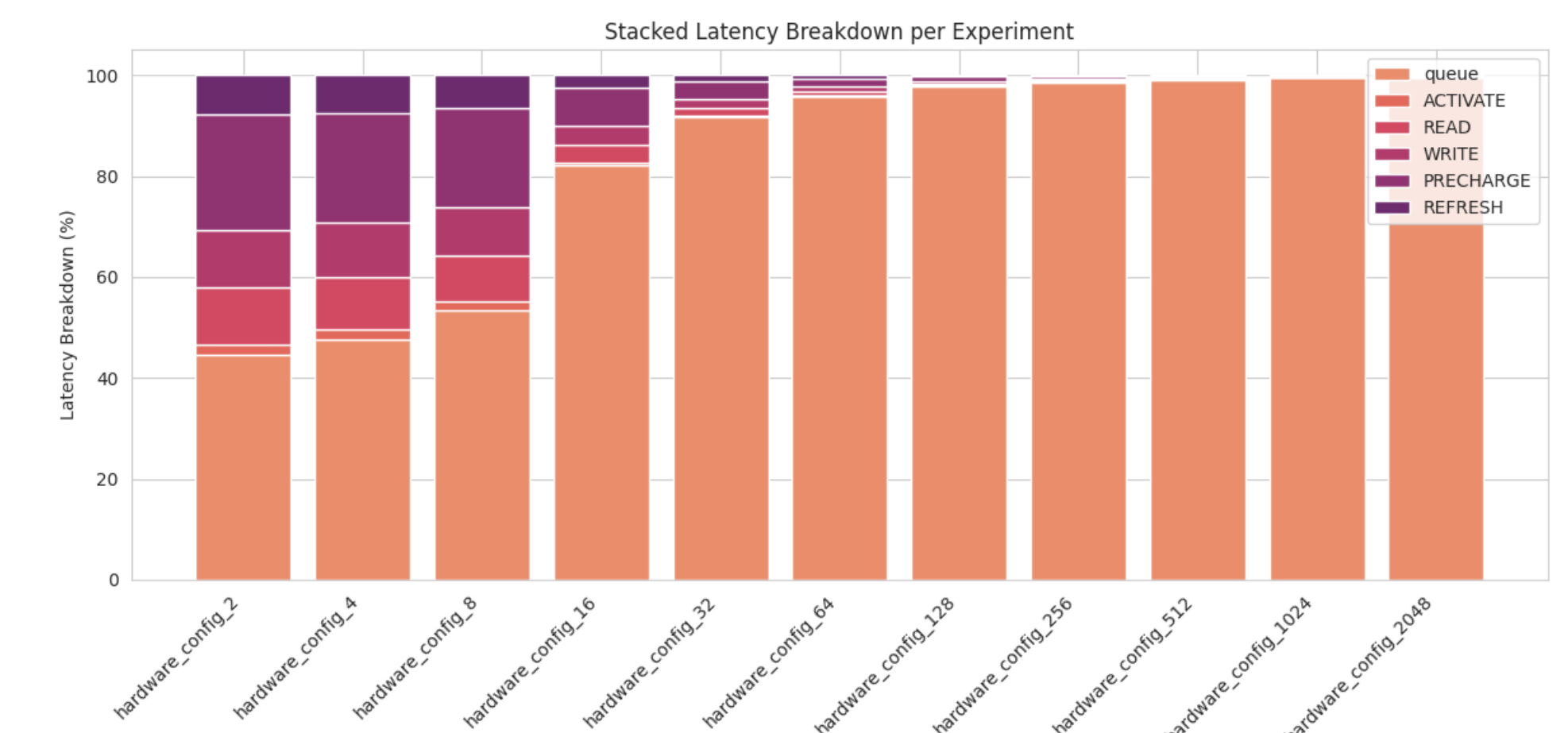


Figure 4: Command-level breakdown of the request latencies as a function of our queue size. We identify **queueSize** as a critical parameter that defines performance in the system.

- Request level-profiles of unified command queue activity

Next steps

- SoC Level FPGA Emulation
 - Leverage FireSim to obtain cycle accurate models of memory performance, area, and energy estimates
- JEDEC HBM Specification Alignment
 - Identify critical features of HBM memories as specified in JEDEC spec, and integrate into memory simulator.
 - Implement application focused bank scheduling policies.
- AI Workload Analysis and Testing
 - Can we identify accelerator level bottlenecks within a hardware system for a given inference workload?

Conclusion

- New software tooling offers stand-alone, high fidelity memory system testing and integrated simulation support.